VOLLEY-G5-ACR1

Datum, tolerance-budget and assembly inspection notes

OUTLINE / PRELIMINARY ENGINEERING — dimensions in mm; no general manufacturing tolerance assigned

PROJECT REFERENCE DATUMS (preliminary; no supplier datum scheme assigned)

A: generic mounting plane Z = -150. B: launch direction +X. C: launcher median plane Y = 0.

Dimensions are basic CAD nominal values. A general ± tolerance or GD&T release is not assigned.

MAGNET/PACKAGE INDIVIDUAL-FACE STACK — declared allocation, not a measured fit

Nominal 1.000 − FE paired closure 0.371778/2 − guide centring 0.200 − flatness 0.150

− assembly 0.150 − differential thermal 0.050 = 0.264111 mm residual per face.

FE closure uses the declared 130% load screen; load/material/mesh limits remain in the source report.

CRADLE ENDWALL STACK — each nominal axial gap 0.500

Machining allocation 0.100 + docking allocation 0.100; differential CTE assumption 12 µm/(m·K).

340.5 mm reference length: residual 0.300000 / 0.218280 / 0.136560 / 0.054840 mm

at differential temperature 0 / 20 / 40 / 60 K. Elastic contacts need later correlation.

ASSEMBLY AND INSPECTION SEQUENCE

1 Identify each part against BOM_HOME.csv and frozen native labels; verify material class and mass assignment.

2 Establish project mounting and rail datums; inspect rail straightness/parallelism within the allocated stack.

3 Assemble insulated stator and sled; inspect both face gaps over home-to-release travel before energizing.

4 Verify cassette guide/channel clearance, screw support alignment and each of twelve prescribed transfer routes.

5 Land pallet; inspect caps, lockpins, both bayonet/grip overlaps and capture-before-release order.

6 Inspect cam opening/running clearances and full cradle withdrawal; execute powerless/manual custody checks.

7 Record as-built dimensions, exceptions and inspection instrument calibration before a hardware test release.

Acceptance checklist: INSPECTION_PLAN.csv. No inspection, material certification, torque or physical test is recorded here.